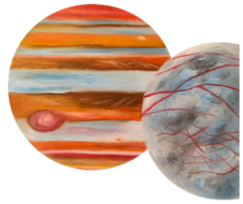
Caroline Haslebacher
Curriculum Vitae



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https://carolinehaslebacher.github.io/

PROFESSIONAL EXPERIENCE

01/2025 — today **Postdoctoral Research Fellow at Southwest Research Institute (SwRI), Boulder**

02/2026 — today **Contractor at Laboratory for Atmospheric and Space Physics, University of Colorado, Boulder**

10/2024 — 12/2024 **Postdoctoral Researcher at Department of Space Research & Planetary Sciences, Physics Institute, University of Bern**

EDUCATION

10/2020 — 09/2024 **PhD in Physics, Department of Space Research & Planetary Sciences, Physics Institute, University of Bern**
"Reading between the lines: Linear surface features on Jupiter's moon Europa interpreted with Deep-Learning", supervised by Prof. Dr. Nicolas Thomas

09/2019 — 09/2020 **Master of Science in Physics, University of Bern**
"The Impact of Climate Change On Site Characteristics of Extremely Large Telescopes", supervised by Prof. Dr. Brice-Olivier Demory

09/2016 — 09/2019 **Bachelor of Science in Physics, University of Bern**
"Polarization gating for imaging subsurface structures in backscattering", supervised by Prof. Dr. Martin Frenz

MISSION INVOLVEMENT - NASA EUROPA CLIPPER

03/2025 — today **Co-Leader of the Europa Clipper Sunrise Group**

05/2021 — today **Affiliate of the Europa Imaging System (EIS) instrument science team**
Lead of the EIS target database development
Development of a photometrically corrected USGS basemap for Europa

MENTORING

06/2026 — 08/2026 **Europa Clipper ICONS mentor of intern Katelyn Foley**
"Compositional Analysis of Linear Surface Features"

01/2026 — today **Mentor for APS independent study of Jesse Magni, co-supervised with Prof. Paul Hayne**
"Geologic age sequencing of linear surface features on Jupiter's moon Europa"

06/2025 — 08/2025 **Europa Clipper ICONS mentor of intern Elise Olmstead**
"MicroMapper: a Deep-Learning Tool for Microfeatures on Europa"

TEACHING

02/2020 — 12/2024 **Teaching assistant, University of Bern**
Physics for veterinary students
Physics for medical students

09/2015 — 09/2020 **Private tutor in mathematics and physics, Bern**
Tutoring of primary, secondary, high-school, and university-level students
Students preparation for high-school entrance exam

AWARDS & PRIZES

03/2026 **Anton Fink Science Prize for Artificial Intelligence (€5,000)**
Doctoral Thesis Prize

05/2023 **Outstanding Student and PhD candidate Presentation (OSPP) Awards 2023**
"Mapping linear surface features on Europa using a deep learning framework" (C. Haslebacher, N. Thomas, EGU23)

GRANTS / FELLOWSHIPS

06/2024 **Postdoc.Mobility fellowship (2 years), Swiss National Science Foundation ($145,276)**
"Unravelling Europa's tidal evolution through lineament sequencing" (with Dr. Alyssa Rhoden)

12/2023 **MERAC Travel Award (Mobilising European Research in Astrophysics and Cosmology), Swiss Society for Astrophysics and Astronomy ($4,700)**
"Advancing the science strategy for the Europa Imaging System"

OUTREACH

12/2024 **Visit to a Kindergarten, Eichberg, Switzerland**
Development of a lesson about Europa with interactive elements in collaboration with Richa Wüthrich

05/2024 **Pint of Science Talk, Brüw (Brewpub), St. Gallen, Switzerland**

05/2023 **Pint of Science Talk, O Bolles Cafe and Bar, Bern, Switzerland**

01/2021 **Primary school visit**
Anaglyphs of Mars and Europa prepared in booklets

SERVICE

02/2025 — today **Co-Leader of the AI@SwRI group together with Dr. Andrés Munoz**
A group I created to upskill together, touching on all aspects of Artificial Intelligence

2022 — today **Referee for Advances in Space Research, Nature (Scientific Reports), JGR Planets**

01/2024 — 12/2024 **Member of the Colloquium Organizing Committee at the Department of Space Research & Planetary Sciences at the University of Bern**

03/2022 — 07/2022 **Swiss Institute for Planetary Sciences (SIPS) working group 'Vision and Research until 2050'**
Responsible for small bodies (up to Earth mass) in our solar system

INVITED SEMINAR TALKS

04/2026 **Deggendorf Institute of Technology**

03/2026 **Stony Brook University, Department of Geosciences**

05/2025 **Jet Propulsion Laboratory / California Institute of Technology**

01/2024 **Southwest Research Institute, Boulder**

EXTRACURRICULAR ACTIVITIES

09/2015 — 07/2016 **Study of music, University of the Arts (HKB), Bern**
Viola with Prof. Getrud Weinmeister

07/2018 — 02/2020 **Intern Surgical Navigation (30%), CAScination AG, Bern**
Development of tissue-mimicking phantoms with ultrasound- and computed-tomography-characteristics
Verification & Validation of medical software

CAROLINE HASLEBACHER
PUBLICATION LIST

01. May 2026

Total number of published papers	8
1st-author papers	3
2nd-author papers	3
1st-author book chapters	2
1st-author datasets	4
Number of citations	93 (Google Scholar) / 53 (Scopus)
H-index	4 (Google Scholar and Scopus)

PEER-REVIEWED PUBLICATIONS (8)

05/2026 Peixoto, V. F., Morgado, B. E., Tognoli, F. M. W., **Haslebacher, C.** (2026). *Topological Analysis of Europa's Fracture Networks: Quantitative evidence for Hemispheric Asymmetry*. Icarus, 449(116943). https://doi.org/10.1016/j.icarus.2026.116943

11/2025 Dümbgen, L., **Haslebacher, C**. (2025). *Nonparametric Smoothing of Directional and Axial Data*. Statistica Neerlandica 79, e70014. https://doi.org/10.1111/stan.70014

05/2025 **Haslebacher, C.**, Tejero, J. G., Prockter, L. M., Leonard, E. J., Rhoden, A. R., Thomas, N. (2025). *Length, Width, and Relative Age Analysis of Lineaments in the Galileo Regional Maps with LineaMapper*. The Planetary Science Journal, 6(6), 133. https://doi.org/10.3847/PSJ/ADD349

12/2024 Turtle, E., McEwen, A., Patterson, G., Ernst, C., Elder, C., Slack, K., McDermott, J., Meyer, H., DeMajistre, R., Espiritu, R., Bland, M., Becker, M., Collins, G., Corlies, P., Darlington, H., Derr, C., Daubar, I., Detelich, C., Edens, W., Fletcher, L., Gardner, C., Graham, F., Hansen, C., **Haslebacher, C.**, Hayes, A., Humm, D., Hurford, T., Kirk, R., Kutsop, N., Lees, W., Lewis, D., London, S., Magner, A., Mills, M., Barr Mlinar, A., Morgan, F., Nimmo, F., Ocasio Milanes, A., Phillips, C., Pommerol, A., Prockter, L., Quick, L., Robbins, G., Stickle, A., Stewart, B., Sutton, S.S., Thomas, N., Torres, I., Tucker, O., Auken, V., Wilk, K., Seifert, H., Niewola, J. (2024). *The Europa Imaging System (EIS) Investigation*. Space Science Reviews, Collection " Europa Clipper: A Mission to Explore Ocean World Habitability," 220. https://doi.org/10.1007/s11214-024-01115-9

03/2024 **Haslebacher, C.**, Thomas, N., Bickel, V. T. (2024). *LineaMapper: A deep learning-powered tool for mapping linear surface features on Europa*. Icarus, 410(115722). https://doi.org/10.1016/j.icarus.2023.115722

09/2022 **Haslebacher, C.**, Demory, M. E., Demory, B. O., Sarazin, M., Vidale, P. L. (2022). *Impact of climate change on site characteristics of eight major astronomical observatories using high-resolution global climate projections until 2050 - Projected increase in temperature and humidity leads to poorer astronomical observing conditions*. Astronomy & Astrophysics, 665, A149. https://doi.org/10.1051/0004-6361/202142493

05/2020 Eigl, B., **Haslebacher, C.**, Müller, P. C., Andreou, A., Gloor, B., Peterhans, M. (2020). *A multimodal pancreas phantom for computer-assisted surgery training*. IEEE Open Journal of Engineering in Medicine and Biology, 1, 166–173. https://doi.org/10.1109/OJEMB.2020.2999786

04/2020 Müller, P. C., **Haslebacher, C.**, Steinemann, D. C., Müller-Stich, B., Hackert, T., Peterhans, M., Eigl, B. (2020). *Image-guided minimally invasive endopancreatic surgery using a computer-assisted navigation system*. Surgical Endoscopy. https://doi.org/10.1007/s00464-020-07540-5

DATASETS (4)

03/2025 **Haslebacher, C.** (2026). *Subsolar Ground 'Sun' Azimuth Galileo images of Europa processed with USGS isis*. Mendeley Data, V1, https://doi.org/10.17632/8n6zvy8vdd.1

06/2025 **Haslebacher, C.**, Gamazo Tejero, J., Prockter, L.M., Leonard, E.J., Rhoden, A.R., Thomas, N. (2025). *Length, width, and relative age analysis of lineaments in the Galileo regional maps with LineaMapper (Haslebacher et al., PSJ, 2025) accompanying data*. Mendeley Data, V2, https://doi.org/10.17632/rjhsjrnxgv.2

06/2025 **Haslebacher, C.**, Thomas, N., Bickel, V.T. (2023). *LineaMapper: A deep learning-powered tool for mapping linear surface features on Europa (Haslebacher et al., Icarus, 2023) accompanying data*. Mendeley Data, V1, https://doi.org/10.17632/nxnwj6hy4s.1

01/2023 **Haslebacher, C.**, Demory, M.-E., Demory, B.-O., Sarazin, M., Vidale, P. L. (2023). *Model agreement and trend analysis data associated to the publication: "Impact of climate change on site characteristics of eight major astronomical observatories using high-resolution global climate projections until 2050"*. Zenodo, https://doi.org/10.5281/zenodo.7541457

BOOK CHAPTERS (4)

04/2025 **Haslebacher, C.**, Bolmont, E., Cilibrasi, M., Grone, J., Haslebacher, N., Helled, R., Kervazo, M., Ligterink, N. F. W., Lovis, C., Mayer, L., Obersnel, L., Ottersberg, R., Oza, A. V., Patty, C. H. L., Pommerol, A., Portyankina, G., Rhoden, A. R., Schlarmann, L., Shibaike, Y., Singh, V., Vorburger, A. H., Wurz, P. Active moons in our Solar System and beyond - Europa, Enceladus, Triton, and (exo)-Io. In SSR collection and book "The National Center for Competence in Research, PlanetS: A Swiss-wide network expanding planetary sciences". Editors: Benz, W., Schönbächler, M., Thomas, N., Udry, S., https://arxiv.org/abs/2604.12104

04/2025 Davoult, J., Bickel, V., **Haslebacher, C.**, Alibert, Y., Gruchola, S., Cantero, C., Egger, J. A., Eltschinger, R., Eyholzer, Y., Garvin, E. O., Gruchola, S., Leleu, A., Marques, S., Zhao, Y. *Machine Learning as a Transformative Tool for (Exo-) Planetary Science*. In SSR collection and book "The National Center for Competence in Research, PlanetS: A Swiss-wide network expanding planetary sciences". Editors: Benz, W., Schönbächler, M., Thomas, N., Udry, S., https://arxiv.org/abs/2604.09152

12/2025 Prieto-Ballesteros, O., Tosi, F., McKinnon, W. B., Rhoden, A. R., Schenk, P., Vance, S. D., Wagner, R., Hao, J., Finkel, P. L., de Dios Cubillas, A., **Haslebacher, C.**, Ding, M. *Geology and Surface Properties of the Galilean Moons*. In SSR collection and ISSI book "Exploring the Jovian Satellite System: From Formation to Habitability". Editors: Blanc, M., Bolton, S., Ding, M., Hao, J., Kimura, T., Lainey, V., Li, L., Nimmo, F., Prieto-Ballesteros, O., Tosi, F., https://link.springer.com/article/10.1007/s11214-026-01273-y

Submitted 06/2024 **Haslebacher, C.**, Thomas, N., Rossi, C., Leonard, E. J., Prockter, L. M., Sabbeth, L., Hasnain, Z., and Tejero, J. G. *A deep-learning powered global lineament map of Europa and applications to other planetary bodies*. In Europlanet book "Atlas of Geological Planetary Mapping". Editors: Marinangeli, L., and Pantaloni, M.

RECENT CONFERENCE PRESENTATIONS

12/2025 **Haslebacher, C.**, Dümbgen, L., Rhoden, A., Leonard, E.J., Prockter, L.M., Thomas, N. (2025). *Lineament Azimuths and Europa's Tidal Stress Field - A Correlation Study*. AGU25.

08/2025 **Haslebacher, C.**, Rhoden, A. R., Ferguson, S. N., Almeida, M., Turtle, E. P., Leonard, E. J., Read, M. (2025). *Optimizing the Uranus Orbiter and Probe Mission with a Prioritized Geological Target Database – An Initial Concept*. International Workshop on Instrumentation for Planetary Missions - 6.

03/2024 **Haslebacher, C.**, Rhoden, A.R., Thomas, N. (2024). *Constraining Lineament Formation on Europa: An Investigation of the Spatiotemporal Evolution of Lineament Azimuths*. 55th Lunar and Planetary Science Conference.